



Dependence of the photocatalytic reduction of bicarbonate to formic acid by Au–TiO₂ on Au morphology and its plasmonic vibrational mode

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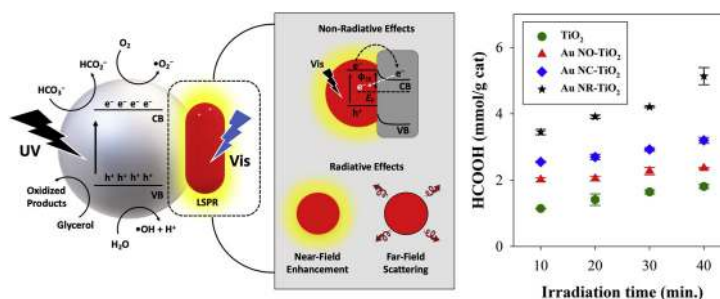
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HIGHLIGHTS

- Au nanoparticles with various shapes were prepared and integrated to TiO₂.
- The role of plasmonic modes in Au–TiO₂ was studied for bicarbonate reduction.
- Photoactivity of Au–TiO₂ was affected by Au plasmonic vibration mode.
- Plasmon-enhanced activity was due to radiative and non-radiative effects.
- Au nanorod–TiO₂ was the most active catalyst for bicarbonate reduction.

GRAPHICAL ABSTRACT



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ABSTRACT

The effect of Au morphology and its plasmonic vibration mode on the photoactivity of Au–TiO₂ nanostructures was evaluated. Here, Au nanoparticles with three distinct morphologies, i.e., nanooctahedra, nanocube, and nanorod, were successfully prepared and integrated into TiO₂ nanoparticles. The nanocatalysts were used for photocatalytic conversion of bicarbonate to formic acid for solar fuel generation. Based on the result, it is found that the integration of Au and TiO₂ resulted in a synergistic effect on the production of formic acid due to the presence of Au's SPR phenomenon. Results also demonstrated that the shape of Au nanoparticles plays a pivotal role in the improvement of Au–TiO₂ activity. From the photocatalytic experiment, Au NR–TiO₂ was found to be the most active nanocatalyst with a formic acid yield of 5.14 mmol/g cat. By using spectroscopy techniques and near-field enhancement simulation, it is believed that the photocatalytic enhancement was due to non-radiative (hot electron transfer and storage) and radiative (near-field enhancement and far-field scattering) effects. However, the additional shape-dependent variation on activity was most likely due to the radiative effects instead of non-radiative ones.

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1. Introduction

The depletion of fossil fuels and their negative environmental impact has driven researchers to seek alternative renewable energy resources. One of the leading technological breakthroughs for solving this issue is the utilization of carbon dioxide (CO_2) in solar fuel generation [1–3]. Among different types of solar fuels, photoconversion of CO_2 to formic acid has recently attracted much attention as it can serve as precursors in the synthesis of many chemicals, hydrogen storage materials, reducing agents, and as the reactant in formic acid-based fuel cell [4]. However, the large-scale operation is still limited by several challenges due to the high thermodynamic stability of CO_2 , making its photoreduction is a difficult and energetically unfavorable process [5–9]. Besides, several practical drawbacks can also be originated from the low CO_2 solubility in water (0.033 M at T: 25 °C and P: 1 atm), which results in the lack of productivity and selectivity for formic acid [10]. Nevertheless, this issue can be avoided by using bicarbonate ion as the reactant as an alternative to the utilization of CO_2 . Typically, CO_2 can easily be captured in the form of NaHCO_3 using various chemisorption materials such as sodium metasilicate (Na_2SiO_3) or encapsulate liquid sorbent [11,12]. A recent study by Leonard and co-workers have suggested that photoreduction of bicarbonate was found to be more favorable than that of CO_2 [13]. It is reported that bicarbonate is more favorable to form coordination with active catalytic centers on the surface of TiO_2 , and thus facilitating a more straightforward reduction [14,15]. Therefore, the development of suitable catalytic materials for the photoreduction of bicarbonate is desired.

Currently, TiO_2 has been widely used as a catalyst in the photoreduction of bicarbonate due to its excellent chemical stability, cost-effectiveness and environmental friendliness [5,16]. The mechanism of TiO_2 in facilitating the reaction is based on the photogeneration of charge carriers, i.e., electrons and holes, upon the absorption of photon energy from the light source [17]. Typically, the photogenerated electrons may be used to reduce the available bicarbonate ions, while the holes are scavenged by the sacrificing donors [18,19]. However, the application of TiO_2 for large-scale solar fuel operation remains challenging, partly because of the mismatch between the solar light energy (mainly at visible region) and the excitation energy of TiO_2 (bandgap energy, E_g : ~3.2 eV) that lies in the UV region [20,21]. In addition, fast recombination of photogenerated electrons and holes was also reported to be responsible for further reduction in photocatalytic efficiency.

Over the last several years, various strategies have been employed to enhance the catalytic activity of TiO_2 and Recently to avoid the occurrence of the previously mentioned shortcomings. For instance, different types of photosensitizers, such as organic dyes, transition metal complexes, or small bandgap semiconductors such as quantum dots, have been used to widen the light absorption spectrum of TiO_2 to match solar radiation [22–24]. It was reported that, upon solar irradiation, these photosensitizers could inject their photogenerated electrons to the conduction band (CB) of TiO_2 , resulting in an increase in the population of hot electrons essential for CO_2 conversion. In other studies, integration with plasmonic particles, like Au or Ag nanoparticles, could also be used to enhance TiO_2 photocatalytic activity [25,26]. Literature suggested that surface plasmon resonance (SPR) of plasmonic particles was able to not only broaden the light absorption ability to visible light region but also to prevent fast charges recombination [27,28]. In our previous study, we reported that the plasmon-induced enhancement of TiO_2 photocatalytic activity was primarily due to several synergistic radiative and non-radiative effects, i.e. (1) SPR-generated hot electrons migration to TiO_2 's CB, (2) generation of intense local electric field “hot-spot” near TiO_2 -plasmonic particles interface, and (3) the ability of plasmonic particles to act as the storage for photogenerated electrons, and thus preventing fast charges recombination [29]. Nevertheless, very few studies have been conducted to investigate the effect of plasmonic vibration on the photocatalytic enhancement of TiO_2 . In many reports, it has been widely reported that the vibration modes of plasmonic

nanoparticles, especially Au, were significantly affected by the size and morphology of the particles [30,31].

In this study, the role of Au nanoparticle's shape to the photocatalytic activity of nano Au- TiO_2 heterostructures was studied. Here, the effect of plasmonic vibration modes was evaluated by fabricating the nano-heterostructures, where Au nanoparticles were made in three different morphological features (nanooctahedra, nanocube, and nanorod). In this work, the as-prepared nano-heterostructures were used as catalysts for the photoreduction of bicarbonate to formic acid. Furthermore, the proposed catalytic mechanism was also studied using both experimental and theoretical approaches to provide comprehensive information in the mechanism of as-prepared nanocatalyst that facilitates the reduction of bicarbonate to formic acid.

2. Experimental section

2.1. Materials

Gold (III) chloride hydrate ($\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$) (purity: >99.5%), sodium citrate (purity: >99%), L-ascorbic acid (purity: >99%), sodium borohydride (NaBH_4) (purity: >98%), hexadecyltrimethylammonium bromide (CTAB) (purity: > 98%) and silver nitrate (AgNO_3) (purity: >99%) were purchased from Sigma-Aldrich and used for the synthesis of Au nanoparticle with different morphological features. For the synthesis of anatase TiO_2 nanoparticles, titanium (IV) butoxide ($\text{Ti}(\text{OBu})_4$) (purity: >97%), NH_4OH solution (28–30% NH_3 in H_2O) and absolute ethanol were obtained from Sigma-Aldrich and used without further purification. Additionally, hydrochloric acid (HCl) were also purchased from Sigma-Aldrich and used in the fabrication of the nanocatalysts. Finally, sodium bicarbonate (NaHCO_3), methanol, formic acid (HCOOH) and potassium dihydrogen phosphate (KH_2PO_4) were purchased from Sigma-Aldrich and used in photocatalytic experiments.

2.2. Synthesis of Au nanooctahedra (Au NO)

In this work, Au NO was prepared via hydrothermal method [32]. Here, monodisperse Au nanooctahedra was synthesized by making an aqueous solution containing 0.055 g of CTAB, 50 μL of 0.1 M sodium citrate, 250 μL of 0.01 M $\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$ and 9.7 mL of deionized water. The mixture was then heated at 110 °C for 6 h for the reaction to take place. After the reaction, the mixture was then cooled to room temperature and centrifugated at 6000 rpm for 20 min to get rid of excess CTAB residue. The precipitate was then collected and re-dispersed in deionized water for further characterizations.

2.3. Synthesis of Au nanocube (Au NC)

A seed-mediated method reported by Liu and co-workers was used to prepare the Au NC [33]. Initially, seeds Au nanoparticles were prepared by adding 0.6 mL of 10 mM cold NaBH_4 into a 7.8 mL mixture of 0.25 mM $\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$ and 100 mM CTAB while being mixed using a magnetic stirrer for 1 h at 37 °C. Furthermore, a growth solution was prepared by mixing a solution containing 0.8 mL of 10 mM $\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$, 6.4 mL of 100 mM CTAB, and 31.2 mL of deionized water into 3.8 mL of 100 mM ascorbic acid. To obtain the final Au NC, 0.02 mL of diluted seed solutions was then added to the freshly prepared growth solution and mixed with a magnetic stirrer. The final solution was then let to sit overnight and used for further investigations.

2.4. Synthesis of Au nanorod (Au NR)

The preparation for Au NR was carried out with a seed-mediated growth method according to work reported by Nikoobakht and El-Sayed [34]. In this method, Au seeds solution was prepared by adding 0.6 mL of 10 mM NaBH_4 into a 10 mL solution of 0.25 mM $\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$ and 0.1 M CTAB and mixed with a magnetic stirrer for 10 min

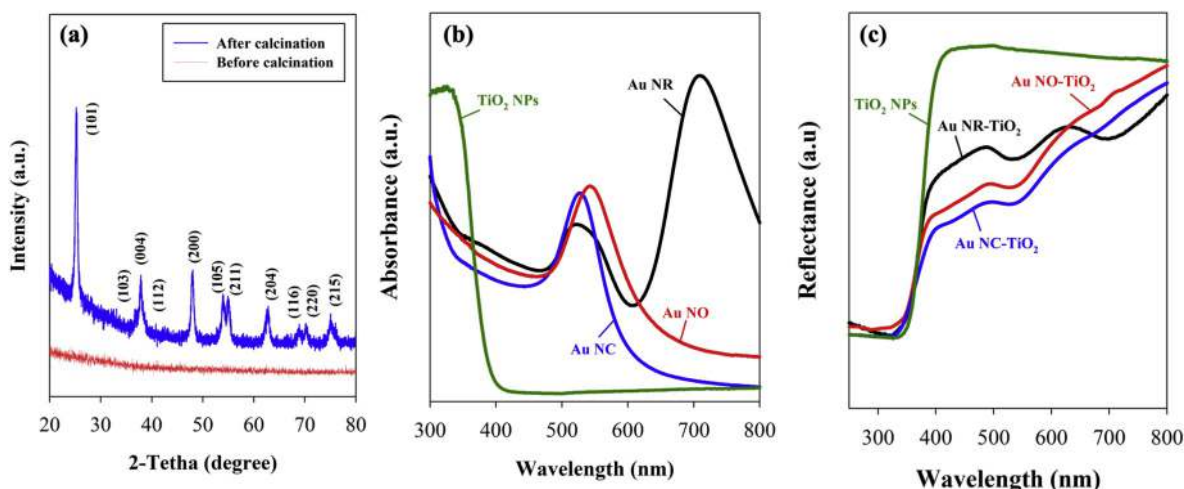


Fig. 1. (a) XRD pattern of the as-prepared TiO_2 nanoparticles (before and after calcination), (b) UV-Vis, and (c) DRS spectra of the as-prepared nanocatalysts.

Meanwhile, the growth solution was prepared by mixing 95 mL aqueous solution containing 0.1 M CTAB, 1 mL of 10 mM AgNO_3 and 5 mL of $\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$. Furthermore, 0.55 mL of 0.1 M ascorbic acid was then added into the solution under vigorous mixing with a magnetic stirrer. In order to prepare the final Au NR, 0.12 mL of the previously made seeds solution was added to the growth solution, and the final mixture was let to sit for 16 h at room temperature and further used for consecutive investigations.

2.5. Synthesis of anatase TiO_2 nanoparticles

The anatase TiO_2 nanoparticles were prepared by the sol-gel method based on Li and co-workers [35]. However, $\text{Ti}(\text{O}i\text{Bu})_4$ was used as the titanium source instead of titanium (IV) isopropoxide (TTIP). In this method, 0.71 mL of $\text{Ti}(\text{O}i\text{Bu})_4$ was added dropwise into a mixture of 0.3 mL of NH_4OH and 100 mL of absolute ethanol while being mixed using magnetic stirrer at 45°C for 3 h. After the reaction, the final mixture was then centrifuged at 6000 rpm for 30 min, and the white precipitates were washed with ethanol and deionized water, respectively. The precipitates were then collected and dried in an oven at 60°C for overnight. Finally, the resulting white powder was then calcined at 450°C for 2 h and used for further characterizations.

2.6. Fabrication of Au- TiO_2 nanocatalysts

In the present work, the as-prepared TiO_2 nanoparticles were integrated with plasmonic Au nanoparticles at acidic condition based on Zhang and co-workers [36]. Here, the 50 mL of each previously prepared colloidal solution of Au nanoparticles were mixed with 0.1 g of the as-prepared TiO_2 nanoparticles. The integration between two nanoparticles was achieved by adjusting the pH of the mixture to pH 3 using 0.01 M of HCl while vigorously stirred using a magnetic stirrer. The mixture was then centrifuged at 6000 rpm for 30 min, and the precipitates were collected and dried in a vacuum oven at 60°C for overnight. Finally, the obtained powder was collected and used for further analyses.

2.7. Characterizations

Several characterization techniques were employed to investigate both the physical and chemical properties of the as-prepared nanocatalysts. To evaluate the crystalline structure of the nanoparticles, X-ray diffraction (XRD) analyses were carried out using PANalytical X'Pert Pro MPD (PANalytical B.V., Amelo, the Netherlands). Here, $\text{Cu-K}\alpha$ radiation was used as the X-ray source while the detection was

carried out using X'Celerator. Furthermore, micrographic images of the as-prepared nanoparticles were also conducted using TECNAI G2 Spirit Twin High-Resolution Transmission Electron Microscope (HR-TEM) to study the morphology, shape, and size of the as-prepared nanocatalysts. Additionally, surface properties of the samples such as Brunauer-Emmett-Teller (BET) surface area, Barret-Joyner-Halenda (BJH) pore volume and BET N_2 adsorption-desorption isotherm were also determined using QUADRASOFT evo (Quantachrome Instruments). Prior to the analyses, 0.1 g of the as-prepared nanocatalysts were firstly degassed using FloVac Degasser. The optical, surface plasmon resonance (SPR) and electronic properties of the nanocatalysts were determined using Shimadzu UV-2450 spectrophotometer. By using the same instrumentation, the optical bandgap (E_g) of the nanocatalysts were determined by carrying out dynamic reflectance spectroscopy (DRS) analyses and estimated using the Kubelka-Munk method (see Supporting Information).

2.8. Photocatalytic experiments

To study the influence of plasmonic vibration modes on the photocatalytic activity of Au- TiO_2 nanocatalysts, photocatalytic reduction of bicarbonate to formic acid was carried out. 25 mL of buffer solution containing 0.3 M NaHCO_3 and 2 M of glycerol as sacrificing donor in deionized water were prepared as the initial reaction mixture. Subsequently, 0.25 mg of each nanocatalyst with different Au morphologies, i. e., nanooctahedra, nanocube, and nanorod, were added into the reaction mixture and stirred in the dark to achieve adsorption equilibrium. Afterward, the photocatalytic reaction was done in a quartz sealed tube and placed under solar simulator with irradiation of light source (150 W Xenon arc lamp, Zolix LSP-X150, China) for 40 min. Here, the horizontal light intensity was approximately 320 cd. An aliquot of the samples was collected at every 10 min increments, and the concentration of formic acid was determined using high-performance liquid chromatography (HPLC). The analyses were performed using Agilent 1200 HPLC system with ZORBAX SB RRHT C18 column and UV detector. For the determination of formic acid, a mixture of 0.1% of KH_2PO_4 and methanol (95:5) was used as the mobile phase, and the detection was conducted at $\lambda = 220$ nm. Furthermore, quantification of formic acid concentration was done by comparing the peak area of the sample with a calibration curve made from standard formic acid solutions (see Supporting Information).

2.9. Near-field enhancement simulation

In this study, simulation for the near-electric field mapping was also

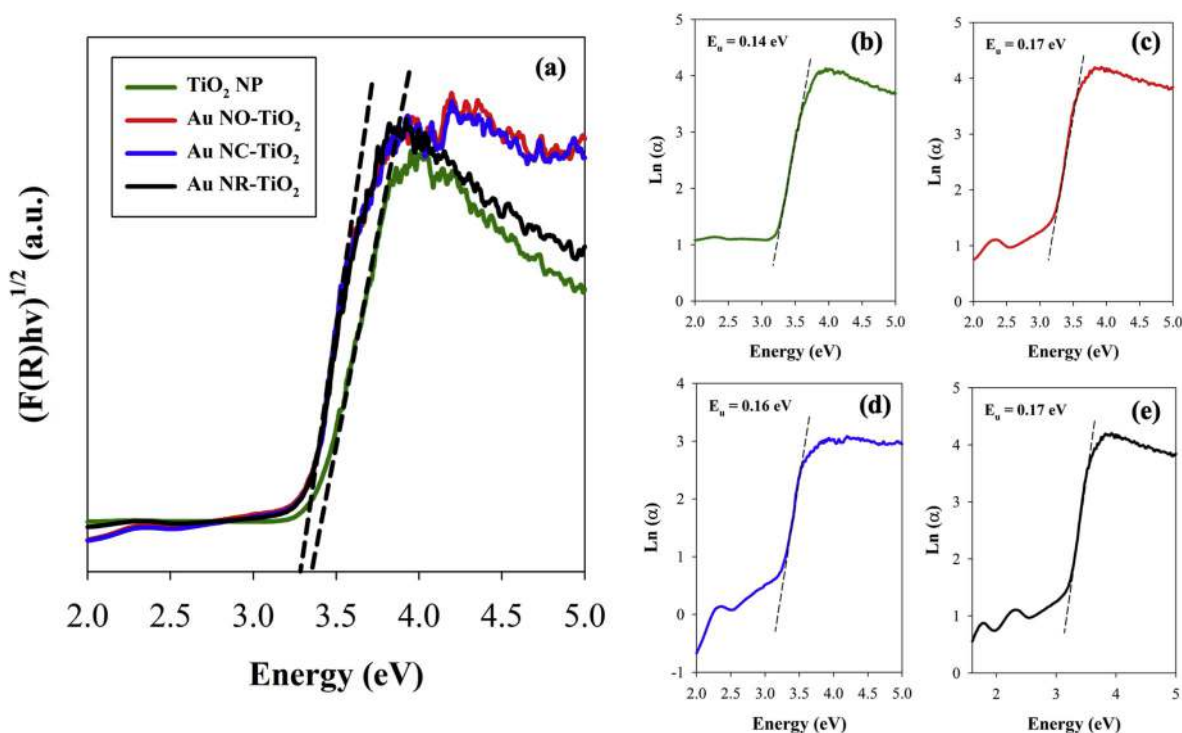


Fig. 2. (a) Kubelka-Munk plot and the corresponding Urbach plots of (b) TiO₂ nanoparticles, (c) Au NO-TiO₂, (d) Au NC-TiO₂ and (e) Au NR-TiO₂.

carried out to further investigate the shape-dependent enhancement in photoactivity due to the radiative effects. Here, the near-electric field map was carried out by electrodynamic simulation packages such as Metallic Nano-Particle Boundary Element Method (MNPBEM) using Boundary Element Method approach [37,38]. Here, the simulation was conducted in water as the medium (refractive index (n): 1.333) in order to mimic the real condition. The length and diameter of Au NR were 59 nm and 12 nm, respectively, and the cell size of Au NC and Au NO were set at 10 nm. Meanwhile, TiO₂ nanosphere with a diameter of 300 nm was placed next to the corresponding Au nanoparticles (Fig. S1, Supporting Information). Furthermore, the electric field in two different polarizations, longitudinal mode (LM) and transverse mode (TM), was applied in order to deeply investigate and understand the enhancement of near-electric field between TiO₂ and Au nanoparticle.

3. Results and discussions

3.1. Preparation and characterization of the nanocatalysts

Fig. 1a presents the XRD pattern of the reaction products obtained from the sol-gel reaction. Based on the result, it is evident that the high-temperature calcination process was responsible for the formation of anatase crystal phase from its amorphous state. A similar phenomenon was also reported elsewhere [39,40]. The diffractogram revealed that all the Bragg's peaks for the calcined sol-gel reaction products could unambiguously be indexed as anatase with no other TiO₂ crystal phases, such as rutile and brookite, were observed [41–43]. The formation of this anatase TiO₂ was also supported by the result for UV-Vis spectroscopic analysis (Fig. 1b). According to the UV-Vis spectrum, the as-prepared TiO₂ nanoparticles showed a strong UV light absorption (<400 nm), while exhibiting no light absorption in the visible region. This strong UV absorption is typically the case for TiO₂ anatase where the absorbed photon energy is used for electronic transition from the valence band (VB) to the conduction band (CB) [44]. Additionally, surface area analysis was also revealed that the resulting TiO₂ nanoparticles had a sufficiently large BET surface area of 40.797 m²/g and BJH pore radius (adsorption) of 34.313 Å, which is beneficial for an

efficient photocatalytic site. Moreover, the BET N₂ adsorption-desorption isotherm exhibited a typical type IV hysteresis loop due to the presence of a mesoporous feature. This proclamation is also supported by the BJH pore size distribution (PSD) analysis (see Fig. S2, Supporting Information).

Furthermore, the evidence for the formation of Au nanoparticles could easily be obtained by monitoring the appearance of strong visible light absorption due to the SPR effect in their UV-Vis spectrum, which is typically absent in their bulk or very small particle counterpart [45]. Results suggest all of the as-prepared Au nanoparticles exhibited a strong light absorption in the visible region (Fig. 1b). Nevertheless, it is worth to note that the position and the amount of the observed SPR bands were found to be dependent on the shape of the Au nanoparticles. In literature, many studies have reported that SPR bands of Au nanoparticles are strongly dictated by the size and shape of the particles [46–48]. As shown in Fig. 1b, the octahedra- and cube-shaped Au nanoparticles exhibited a single SPR peak (λ_{SPR}) at 558 and 542 nm, respectively. Meanwhile, two distinct SPR bands at 552 and 727 nm were observed for Au NR. The appearance of these two bands is due to the oscillation of electrons in different plasmonic vibration modes [48, 49]. Fig. 1b shows the strong SPR band at a longer wavelength, which may be the result of the electron oscillation along the long axis or commonly known as the longitudinal band. In contrast, the weak SPR band at a shorter wavelength was obtained due to transversal electron oscillation.

Additionally, spectroscopic analysis was also used to investigate the formation of the as-prepared Au-TiO₂ nanocatalysts further. Fig. 1c shows the DRS spectra of the as-prepared TiO₂ nanoparticles and the corresponding heterostructures obtained from their integration with different shapes of Au nanoparticles. Based on the results, it is apparent that the as-prepared nano Au-TiO₂ heterostructures absorbed both in the UV and visible light region, indicating that Au nanoparticles were successfully integrated with TiO₂ nanoparticles. In addition to the UV absorption (<400 nm), Au-TiO₂ heterostructure samples also showed a strong light absorption at visible region (400–800 nm). A similar combination of Au and TiO₂ light absorption behaviors in Au-TiO₂ heterostructures was also reported elsewhere [29]. It is believed that this broad

Table 1

Bandgap, band structures, and Urbach energies of the as-prepared nanocatalysts.

Nanocatalysts	Bandgap energy (eV) ^a	Valence band (V) ^b	Conduction band (V) ^b	Urbach energy (eV)
TiO ₂ NPs	3.34	3.03	−0.31	0.14
Au NO–TiO ₂	3.31	3.02	−0.29	0.17
Au NC–TiO ₂	3.32	3.02	−0.30	0.16
Au NR–TiO ₂	3.29	3.00	−0.29	0.17

^a Kubelka-Munk equation.^b Butler and Ginley approach (vs. NHE).

panchromatic-like UV–visible light absorption ability was caused by the synergistic effect between large TiO₂'s bandgap energy and Au's SPR phenomenon. Interestingly, it is also worth to note that the λ_{SPR} of the Au–TiO₂ heterostructures was blue-shifted from the corresponding λ_{SPR} of the individual Au nanoparticles. Based on the result, the λ_{SPR} of the Au NO–TiO₂ and Au NC–TiO₂ were shifted to 545 and 530 nm, respectively. Moreover, both of the transverse and longitudinal λ_{SPR} of Au NR–TiO₂ were also shifted to 542 and 706 nm, respectively.

Further investigations were also carried out to study the effect of both SPR phenomenon and the plasmonic vibration modes on bandgap energy and band structures of the as-prepared Au–TiO₂ samples. Based on the bandgap energy estimation using Kubelka-Munk, it is found that the integration of Au nanoparticles resulted in a slight reduction of bandgap energy of TiO₂ (Fig. 2a). However, it is worth to note that the morphology of Au nanoparticles seemed to have little effect on the

magnitude of this energy reduction. A similar phenomenon was also reported elsewhere [29,50]. This lowering in bandgap energy was most likely due to the presence of localized defect states as the result of the integration of Au nanoparticles. It is believed that these defects could produce an absorption tail that extends deep into the forbidden gap and ultimately narrowing the bandgap of TiO₂ [51]. In literature, this absorption tail is commonly known as the Urbach tail, where its energy can easily be estimated as follows [52,53]:

$$\alpha = \alpha_0 \exp (E/E_u) \quad (1)$$

where, α refers to the absorption coefficient, E is the photon energy, and E_u represents the Urbach energy. Hence, the Urbach energy (E_u) could simply be estimated by plotting $\ln(\alpha)$ vs. photon energy (E) and calculate the value of E_u as the slope reciprocal of the linear portion below the optical bandgap. Fig. 2b–d displays Urbach energy plots of the as-prepared nanocatalysts. As expected, the result demonstrated that the value of Urbach energy of TiO₂ was affected by the presence of Au nanoparticles. Based on the estimation, it is clear that the Urbach energy of TiO₂ was slightly lower than the corresponding Au–TiO₂. It is believed that this increase of Urbach tailing energy was due to the addition of oxygen vacancies in TiO₂ as the result of structural disorder induced by the integration of Au nanoparticles [51]. According to Zhong and co-workers, the magnitude of this absorption tailing is proportional to the density of structural defects [54]. Besides, a recent report also suggested that such phenomenon can also be associated with the enhancement of TiO₂'s ability for trapping the exciton due to the

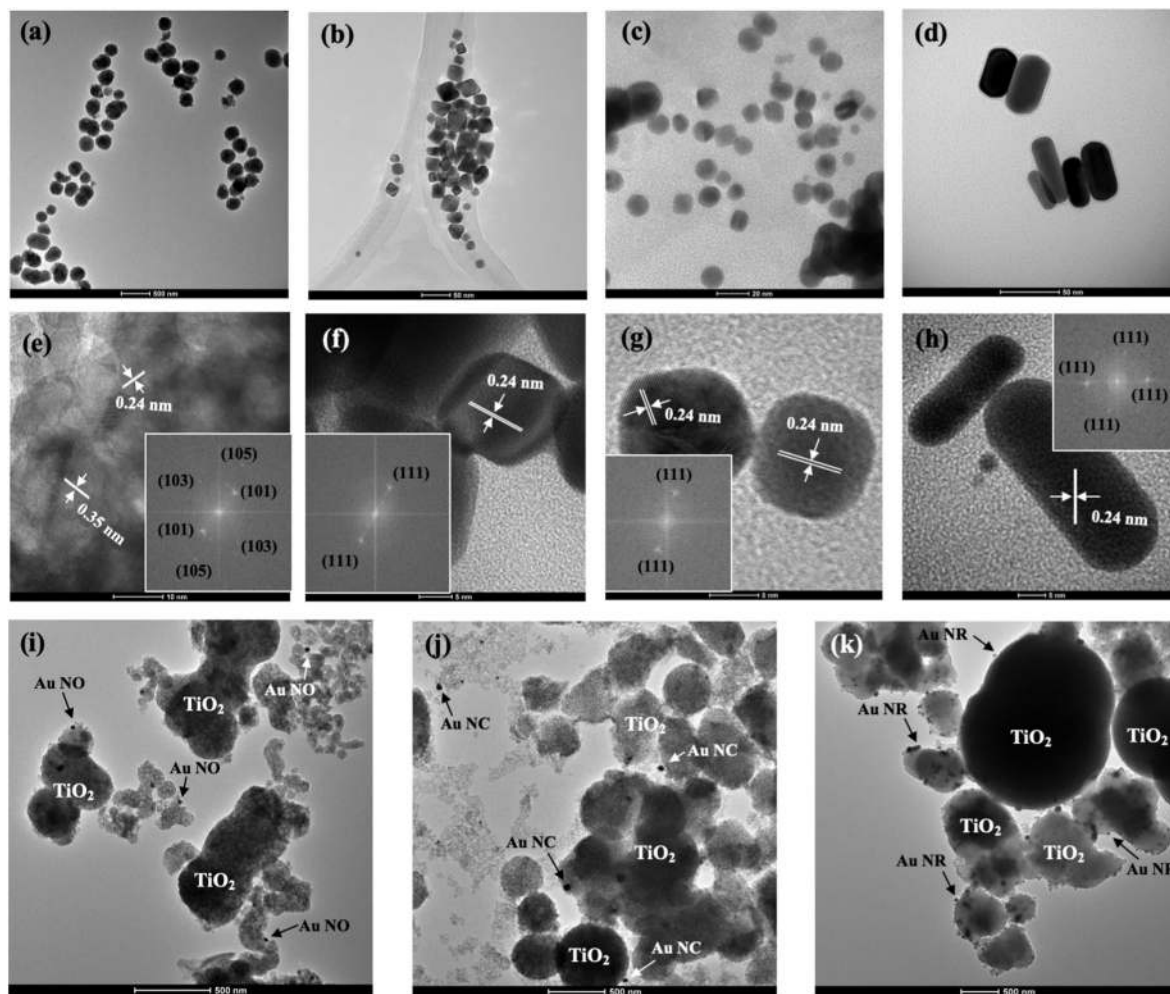


Fig. 3. TEM images of (a) TiO₂ nanoparticles, (b) Au NO, (c) Au NC, (d) Au NR, (e–h) the corresponding HR-TEM images (inset: the FFT analyses), and (i–k) TEM images of the as-prepared nanocatalysts.

synergistic effect from SPR [55].

Based on these calculated bandgap energies, it is also possible to estimate the band structures of the samples using a theoretical prediction approach introduced by Butler and Ginley [56,57]. In this approach, VB and CB energy potential of the samples can be estimated using the following equation:

$$E_{CB} = E^e - X + 0.5(E_g) \quad (2)$$

where E_{CB} is the estimated CB potential, E^e expresses the free electron energy on the hydrogen scale (4.44 ± 0.02 eV), X represents the Sanderson electronegativity of TiO_2 (5.8 eV), and E_g is the calculated Kubelka-Munk band gap energy (eV) [29,58]. Table 1 summarizes the bandgap, band structures, and Urbach energies of the as-prepared pristine TiO_2 nanoparticles and the corresponding Au- TiO_2 nanocatalysts.

An additional evidence for the formation as well as the morphology of the nanocatalysts was also obtained from TEM analyses. Fig. 3 shows the micrographic images of the as-prepared nanoparticles and the corresponding nanocatalysts. Based on the result, it is apparent that the sol-gel reaction followed by high-temperature calcination could be used to fabricate uniform spherical TiO_2 nanoparticles with an average particle size of ~ 300 nm (Fig. 3a). The formation of the anatase crystal phase was further proven from the high-resolution imaging (Fig. 3e). According to result, it is evident that the as-prepared TiO_2 nanoparticles were well faceted with two observable lattice fringes at 0.35 and 0.24 nm, which could be discerned as (101) and (103) lattice planes for anatase, respectively [5,23,25,29,41,43]. This is also supported by the result obtained from the FFT analyses (Fig. 3e, inset).

Furthermore, low magnification TEM observations also revealed that monodisperse Au nanoparticles with octahedral, cubic, and rod shapes were successfully synthesized (Fig. 3b–d). As shown, both Au NO and Au NC exhibited narrow particle size distribution with an average particle size of about ~ 50 nm and ~ 10 nm, respectively. The formation of such well-structured Au morphologies was believed due to the presence of CTAB and sodium citrate as the crystal growth directing agents [32,33]. Investigations with high-resolution imaging and FFT analyses also showed that both Au nanoparticles exhibited the (111) Au face-centered cubic (FCC) lattice planes, which were observed at 0.24 nm lattice fringes (Fig. 3f and g) [31,59]. On the other hand, results also demonstrated that the as-prepared Au NR had an aspect ratio average of ~ 5 with diameter and length of the rod particles of around 12 and 59 nm, respectively (Fig. 3d). The formation of well-faceted Au NR was also supported by the appearance of Au (111) FCC planes in both HR-TEM images and FFT analysis (Fig. 3h). According to Nikoobakht and El-Sayed, the formation of such unique and monodisperse rod-like morphology was believed due to the presence of crystal facet directing agents, i.e., AgNO_3 and CTAB, during Au crystal growth reaction [34]. It is suggested that AgNO_3 and CTAB could react to form AgBr , which has the ability to induce the formation of rod-like particle structures. A detailed crystal growth mechanism for the seed-mediated synthesis of Au NR in the presence of AgNO_3 and CTAB has also been comprehensively reviewed elsewhere [60].

Results from TEM images also demonstrated that the as-prepared Au nanoparticles with different morphological features were successfully incorporated with TiO_2 to form the corresponding Au- TiO_2 nanocatalysts (Fig. 3i–k). Based on the result, the integration of the two materials could easily be identified by the appearance of several small Au nanoparticles with distinct particle shapes on the surface of large sphere-like TiO_2 nanoparticles. Besides, under the electron microscope, Au nanoparticles could easily be differentiated as they typically have a darker contrast than TiO_2 due to their high electron density [61].

3.2. Evaluation of photoreduction of bicarbonate

Photocatalytic experiments for the reduction of bicarbonate were

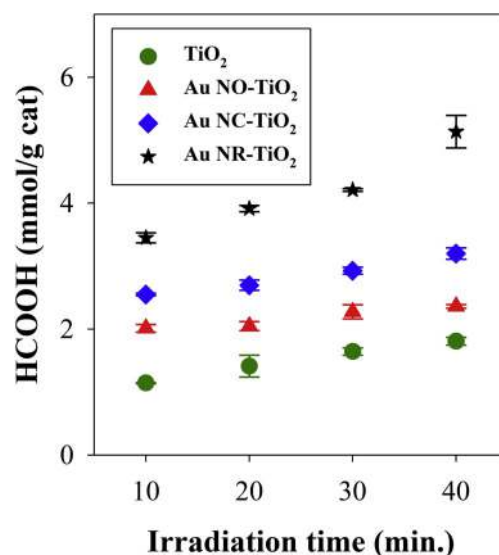


Fig. 4. Production of formic acid using the as-prepared nanocatalysts under artificial solar irradiation.

carried out to evaluate the photocatalytic activity of the as-prepared nanocatalysts. In this work, the performance of each nanocatalyst was assessed by monitoring the resulting formic acid concentration during light irradiation. Fig. 4 shows the performance of the as-prepared pristine TiO_2 and the corresponding Au- TiO_2 nanocatalysts in the photo-reduction of bicarbonate. Based on the result, it is apparent the photocatalytic activity of TiO_2 nanoparticles could be enhanced by integrating Au nanoparticles. Under artificial solar light irradiation at ambient reaction condition, the production of formic acid using different types of Au- TiO_2 nanocatalysts were at least two times greater than the pristine TiO_2 nanoparticles. According to recent studies by Pan and co-workers, the observed photocatalytic enhancement could be attributed due to the synergistic contribution of Au's SPR effect [18,19]. Similar photocatalytic improvement was also obtained when such nanostructures were used in other reactions such as photodegradation of organic dyes/pollutants [23,25,26,29,61].

It is argued that Au nanoparticles can readily absorb visible light energy due to the SPR effect. The photoexcited plasmons can then non-radiatively decay to create an impermanent non-equilibrium electron-hole pair, where the hot electrons can be transferred to the adjacent TiO_2 's CB via quantum tunneling, provided the energy of the hot electrons is larger than the Schottky barrier (SB) height (Φ_{SB}) at the interface between the metal and semiconductor [62–64]. Typically, the energy of these hot electrons is in range of 1–4 eV for Au nanoparticles, while the value of Φ_{SB} in Au- TiO_2 interface was reported to be around 0.15–0.23 eV [65–67]. The evidence for such migrations has also been experimentally investigated using IR probe and femtosecond transient absorption spectroscopy [68,69]. Additionally, this phenomenon can also be noticed through the shift of VB of Au- TiO_2 with respect to the corresponding pristine TiO_2 nanoparticles [70]. It is reported that both electron transfer and the formation of SB can drive the Fermi level into more negative potential causing the position of TiO_2 's VB to shift to the lower binding energy. In this work, the calculated Au- TiO_2 's VB positions were found to be less positive than the pristine TiO_2 's VB (Table 1). Nevertheless, since spectral light absorption overlap between TiO_2 and Au's SPR was absent (Fig. 1b), the plasmon-induced energy transfer cannot be the main reason for the observed photocatalytic enhancement. Besides, the photocatalytic enhancement due to the plasmonic heating under solar irradiation was also irrelevant, considering the high thermal energy required for CO_2 to formic acid conversion (0.61 V) [5, 71].

Takai and Kamat suggested that the ability of plasmonic

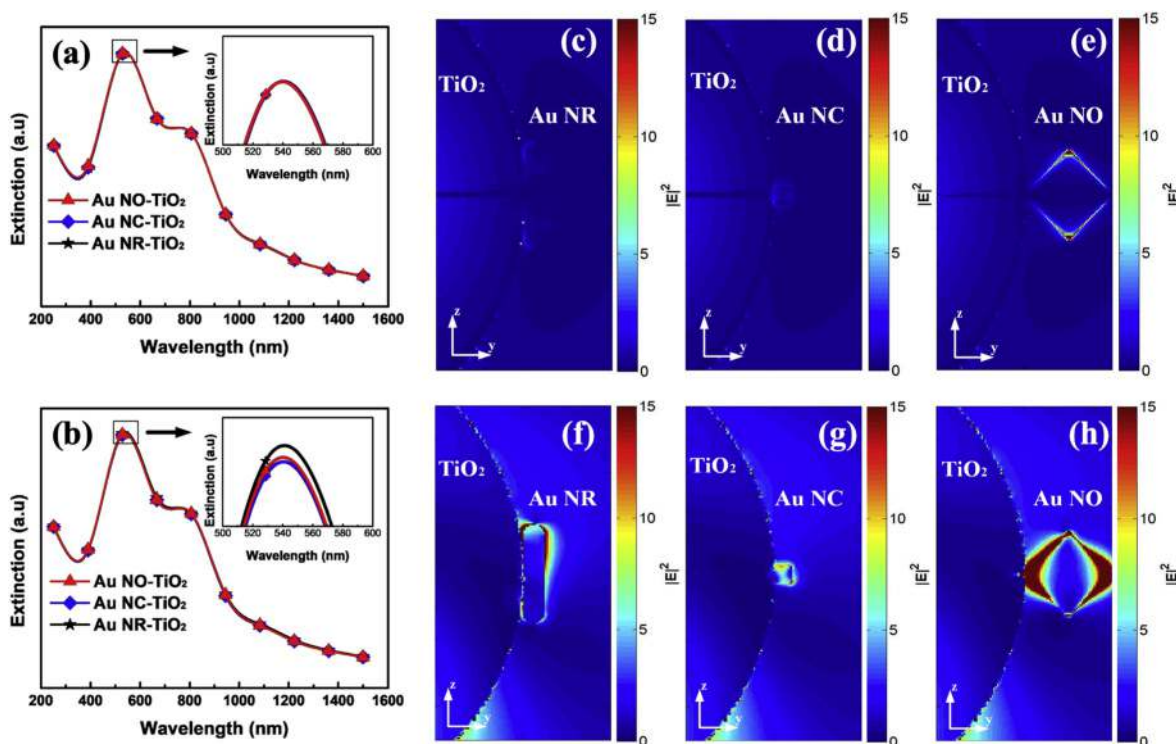


Fig. 5. Far-field scattering extinction due to (a) longitudinal and (b) transverse vibrational modes of the nanocatalysts; and the corresponding near-field enhancement for (c–e) longitudinal and (f–h) transverse modes of Au NR-TiO₂, Au NC-TiO₂, and Au NO-TiO₂, respectively.

nanoparticles to prevent electron-hole recombination could also contribute to the overall enhancement in photocatalytic activity [72]. The prevention of this electron-hole recombination was believed due to the ability of Au nanoparticles to act as a reservoir for the photo-generated hot electron in TiO₂ segment [29,70,72]. Under artificial solar irradiation, it is believed that the hot electrons in TiO₂'s CB can migrate to Au nanoparticles. This migration is possible since the Fermi level of Au is more positive (0.75 V) than the calculated TiO₂'s CB (Table 1). The evidence for this storage phenomenon can easily be identified by a blue-shift in the surface plasmon frequency [70]. According to the result, it is apparent that the λ_{SPR} of nanocatalysts blue-shifted from their corresponding Au's λ_{SPR} (Fig. 1b and c). Interestingly, results also demonstrated that the enhancement rate of photocatalytic activity due to the plasmonic effect was also affected by the shape of Au nanoparticles. As shown, the photocatalytic activity of the as-prepared nanocatalysts could be arranged according to the following: Au NR-TiO₂ > Au NC-TiO₂ > Au NO-TiO₂ > pristine TiO₂. The yield of formic acid was found to be 2.36, 3.20, and 5.14 mmol/g cat when Au NO-TiO₂, Au NC-TiO₂, and Au NR-TiO₂ were used as the nanocatalysts under solar irradiation for 40 min, respectively. Meanwhile, in the same condition, pristine TiO₂ nanoparticles could only produce 1.81 mmol formic acid/g cat.

It is believed that this shape-dependent variation in photocatalytic activity was primarily due to radiative effects rather than non-radiative ones. According to literature, radiative effects can be attributed to the presence of far-field light scattering and near-field enhancement [73, 74]. Here, the presence of this effect was studied using near-field enhancement simulation. Fig. 5 shows the results of both near-field enhancement and far-field scattering for the as-prepared nanocatalysts in longitudinal and transverse vibrational modes. In far-field scattering, it is believed that Au could reflect and scatter a portion of unabsorbed incident light due to their sufficiently high albedo properties. Based on the result, light extinction due to scattering was found to be profoundly affected by the particle size, shape, and the applied vibration mode. In Fig. 5a, the light scattering profile for longitudinal vibration mode was

found to be identical for all of the nanocatalysts. However, when the transverse vibrational mode was applied, Au NR-TiO₂ resulted in a slightly higher extinction peak at 540 nm (Fig. 5b, inset). This suggests that Au NR was able to scatter more light than Au NO and Au NC, which is consistent with the superiority of Au NR-TiO₂ in facilitating the photoconversion of bicarbonate to formic acid (Fig. 4). This result is consistent with Mie's theory, where the amount of light scattering is proportional to particle size and shape [75]. Besides, light scattering can also be affected by the polarizability (the ability to polarize the incoming light) of the particle, which typically dictated by its geometry. In many cases, this polarization is often observed in non-spherical structures with sharp edges such as nanocube and nanooctahedra [76–78].

Furthermore, a similar phenomenon was also observed in near-field enhancement. According to literature, the presence of SPR from Au nanoparticles could induce the formation of intense local electromagnetic field near the surface of Au nanoparticles [79]. This localized field enhancement can induce the formation of “hot-spot” regions, where the rate of charge carriers generation is 1000 times larger than that of the incident electromagnetic field [70]. Therefore, the population of photo-generated electrons and holes is expected to be higher within this region. Based on the result, it is clear that the near-field enhancement was found to be affected by the type of vibrational modes. In the longitudinal mode, the result demonstrated the magnitude of field enhancement was not affected by the variation of Au morphology (Fig. 5c–e). However, the formation of the “hot-spot” region due to near-field enhancement became more pronounced when transverse vibration mode was applied (Fig. 5f–h). As a result, the photocatalytic activity of Au-TiO₂ nanocatalyst was also expected to be higher since the population of hot electrons and holes essential for the photocatalytic reaction was increased. It is also worth to note that the magnitude of near-field enhancement was more prominent in Au NR-TiO₂ and Au NO-TiO₂ due to the better polarizability of Au nanoparticles.

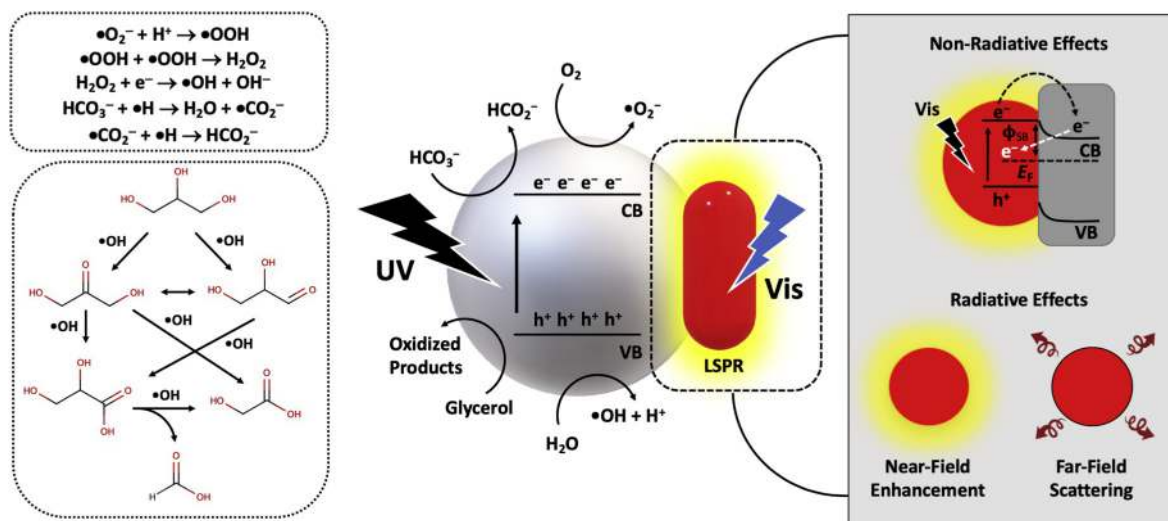
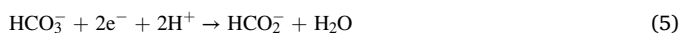


Fig. 6. Schematic representation of photocatalytic bicarbonate reduction to formic acid.

3.3. Mechanism of photocatalytic reaction

Tremendous efforts have been carried out to investigate the mechanism of photoreduction of bicarbonate in the presence of semiconductor-based photocatalysts. Molinari et al. suggested that the formation of $\bullet\text{OH}$ can be used to indicate the occurrence of photocatalytic reduction of bicarbonate in aqueous solution [17]. Using the EPR spin trapping technique, it is reported that the amount of $\bullet\text{OH}$ was found to be proportional to the concentration of bicarbonate and depends on the pH of the solutions. Several relevant reactions responsible for the production of formic acid in the presence of TiO_2 can be expressed as follows:



As shown, $\bullet\text{OH}$ radicals are initially formed from water oxidation (Eq. (4)). This is possible since the photogenerated hole at TiO_2 's VB has more positive potential energy (Table 1) than E° of $\bullet\text{OH}/\text{H}_2\text{O}$ (2.33 V vs. NHE) [17,29]. Meanwhile, bicarbonate can readily scavenge the hot electrons to form formate (Eq. (5)) and useful for enhancing the lifetime of the photogenerated charge carriers. However, bicarbonate can also competitively react with $\bullet\text{OH}$ to form carbonate (Eq. (6)). This could lead to a reduction in formic acid production. In some cases, further suppression in formic acid yield could also be observed since more $\bullet\text{OH}$ radicals are produced during competitive oxygen reductions according to the following reactions [17,29]:



The above reactions are possible since the photogenerated electron in TiO_2 's CB has a more negative reduction potential (Table 1) than E° of $\text{O}_2/\bullet\text{O}_2^-$ (-0.18 V vs. NHE) [80]. Therefore, we used glycerol as a hole scavenger to circumvent these issues. It is believed that glycerol was able to prevent water oxidation (Eq. (4)) due to its exceptional positive hole scavenging ability. Glycerol's superiority as a hole scavenger can be rationalized by the presence of multiple primary OH groups. The OH

groups can facilitate an efficient glycerol's adsorption on the surface of TiO_2 and favor the efficient hole transfer. Besides, this adsorption is beneficial for improving the lifetime of photogenerated charge carriers. Pan et al. reported that the yield of formic acid was found to be proportional to the concentration of glycerol [18]. In other reports, glycerol was also proven to be able to react with the photogenerated $\bullet\text{OH}$ and $\bullet\text{O}_2^-$ and thus possibly repress the occurrence of bicarbonate oxidation (Eq. (4)) [81,82]. It is reported that several oxidation products such as dihydroxyacetone (DHA), glyceric acid (GCA), glycolic acid (GCOA), glyceraldehyde (GCD), hydroxy acetaldehyde (HAA), including formic acid were obtained from glycerol [82].

Furthermore, formic acid can also be possibly obtained from another reaction pathway. It is believed that formic acid can be produced from the reduction of bicarbonate by $\bullet\text{H}$ radicals according to the following reactions:



Here, $\bullet\text{H}$ radicals are believed to be originated from the reduction of H^+ obtained during water oxidation (Eq. (4)). One might argue that such a reaction is unlikely on the surface of TiO_2 . However, when TiO_2 is modified with noble metal nanoparticles, the involvement of $\bullet\text{H}$ in bicarbonate reduction cannot be excluded in this study where Au- TiO_2 was used as the catalysts. As proposed in earlier studies, this phenomenon has been associated with the ability of metal nanoparticles to store hydrogen via the formation of metal-hydride bonds [83,84]. Besides, the presence of $\bullet\text{CO}_2^-$ as intermediate has been experimentally proven using EPR spin trapping analysis [17]. Overall, the proposed plasmon-induced photocatalytic mechanism of bicarbonate reduction to formic acid using Au- TiO_2 is presented in Fig. 6.

4. Conclusion

The performance of the as-prepared Au- TiO_2 nanocatalysts for an enhanced visible-light photoreduction of bicarbonate to formic acid has been successfully investigated in this study. The influence of plasmonic vibration modes on the overall catalytic activity was evaluated by fabricating the nanocatalysts with different Au morphologies, i.e., nanooctahedra, nanocube, and nanorod. Based on the result, the integration of Au nanoparticles to TiO_2 was found to be responsible for the substantial improvement in the production of formic acid, due to the presence of synergistic effects of Au's SPR phenomenon via non-radiative (hot electron transfer and storage) and radiative (near-field

enhancement and far-field scattering) effects. Besides, the shape of Au nanoparticles plays a critical role in the improvement of Au-TiO₂ photocatalytic activity. Au NR-TiO₂ was able to facilitate the production of formic acid by nearly five times the amount produced by pristine TiO₂. Meanwhile, the yield of formic acid was only increased by ~3 times when Au NO-TiO₂ and Au NC-TiO₂ were used as the catalysts. It is believed that the observed shape-dependent enhancement in the photoreduction of bicarbonate was mainly contributed by the radiative effects rather than the non-radiative one. Nevertheless, the contribution from the non-radiative effect due to the plasmon energy transfer from the additional longitudinal band in Au NR-TiO₂ shall not be neglected.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Munawar Khalil: Conceptualization, Supervision, Funding acquisition, Writing - review & editing. **Reza Imam Pratama:** Data curation, Investigation, Software, Visualization. **Muhammad Sujak:** Data curation, Investigation, Software, Visualization. **Alexander Garry:** Data curation, Investigation, Software, Visualization. **Dede Djuhana:** Methodology, Validation, Resources, Data curation. **Aminah Umar:** Methodology, Validation, Resources, Data curation. **Chin Wei Lai:** Methodology, Validation, Resources, Data curation. **Badrul Mohamed Jan:** Methodology, Validation, Resources, Data curation.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matchemphys.2020.123018>.

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